ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

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Project Name _____
Project Number _____

17.5IO Structural Checklist for Building Types S2: Steel Braced Frames with Stiff Diaphragms and S2a: Steel Braced Frames with Flexible Diaphragms

Table 17-11. Immediate Occupancy Structural Checklists for Building Types S2 and S2a.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN AXIAL STRESS CHECK: The axial stress caused by gravity loads in columns subjected to overturning forces is less than $0.10F_y$. Alternatively, the axial stress caused by overturning forces alone, calculated using the Quick Check procedure of Section 4.4.3.6 is less than $0.30F_y$.	5.5.2.1.3	A.3.1.3.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BRACE AXIAL STRESS CHECK: The axial stress in the diagonals, calculated using the Quick Check procedure of Section 4.4.3.4 is less than $0.50F_y$.	5.5.4.1	A.3.3.1.2	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation with a minimum of two anchor rods and with the base plates bearing on concrete or a grout pad. The anchor rods are capable of resisting the overturning force using the Quick Check procedure of Section 4.4.3.6.	5.7.3.1	A.5.3.1	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity (Complete the Following Items in Addition to the Items for Very Low Seismicity)							
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO STEEL FRAMES: Diaphragms are connected for transfer of seismic forces to the steel frames, and the connections are able to develop the lesser of the strength of the frames or the diaphragms.	5.7.2	A.5.2.2	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate Seismicity (Complete the Following Items in Addition to the Items for Very Low and Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of braced frames in each principal direction is greater than or equal to 2. The number of braced bays in each line is greater than 3.	5.5.1.1	A.3.1.1.1	

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C	NC	N/A	U			
☐	☐	☐	☐	COLUMN SPLICES: All column splice details located in braced frames develop 100% of the tensile strength of the column.	5.5.4.2	A.3.3.1.3
☐	☐	☐	☐	SLENDERNESS OF DIAGONALS: All diagonal elements required to carry compression have *Kl/r* ratios less than 200.	5.5.4.3	A.3.3.1.4
☐	☐	☐	☐	CONNECTION STRENGTH: All the brace connections develop the buckling capacity of the diagonals.	5.5.4.4	A.3.3.1.5
☐	☐	☐	☐	OUT-OF-PLANE BRACING: Braced frame connections that are attached to beam bottom flanges located away from beam–column joints are braced out of plane at the bottom flange of the beams.	5.5.4.5	A.3.3.1.6
☐	☐	☐	☐	COMPACT MEMBERS: All brace elements meet compact section requirements in accordance with AISC 341, Table B4.1a.	5.5.4.3	A.3.3.1.7
☐	☐	☐	☐	K-BRACING: The bracing system does not include K-braced bays.	5.5.4.6	A.3.3.2.1
☐	☐	☐	☐	TENSION-ONLY BRACES: Tension-only braces do not comprise more than 70% of the total seismic-force-resisting capacity in structures more than two stories high.	5.5.4.7	A.3.3.2.2
☐	☐	☐	☐	CHEVRON BRACING: Beams in chevron, or V-braced, bays are capable of resisting the vertical load resulting from the simultaneous yielding and buckling of the brace pairs.	5.5.4.6	A.3.3.2.3
☐	☐	☐	☐	CONCENTRICALLY BRACED FRAME JOINTS: All the diagonal braces frame into the beam–column joints concentrically.	5.5.4.8	A.3.3.2.4
Diaphragms (Stiff or Flexible)						
☐	☐	☐	☐	OPENINGS AT FRAMES: Diaphragm openings immediately adjacent to the braced frames extend less than 15% of the frame length.	5.6.1.3	A.4.1.5
☐	☐	☐	☐	PLAN IRREGULARITIES: There is tensile capacity to develop the strength of the diaphragm at reentrant corners or other locations of plan irregularities.	5.6.1.4	A.4.1.7
☐	☐	☐	☐	DIAPHRAGM REINFORCEMENT AT OPENINGS: There is reinforcing around all diaphragm openings larger than 50% of the building width in either major plan dimension.	5.6.1.5	A.4.1.8

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Flexible Diaphragms							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CROSSTIES: There are continuous crossties between diaphragm chords.	5.6.1.2	A.4.1.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 12 ft (3.6 m) and aspect ratios less than 1-to-1 in the direction being considered.	5.6.2	A.4.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 30 ft (9.2 m) and aspect ratios less than or equal to 3-to-1.	5.6.2	A.4.2.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	NON-CONCRETE-FILLED DIAPHRAGMS: Bare steel deck diaphragms or steel deck diaphragms with fill other than reinforced concrete consist of horizontal spans of less than 40 ft (12.2 m) and have aspect ratios less than 4-to-1.	5.6.3	A.4.3.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1	
Foundation System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OVERTURNING: The ratio of the least horizontal dimension of the seismic-force-resisting system at the foundation level to the building height (base/height) is greater than $0.6S_a$.	5.4.3.3	A.6.2.1	
High Seismicity (Complete the Following Items in Addition to the Items for Very Low, Low, and Moderate Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONNECTION STRENGTH: All the brace connections develop the yield capacity of the diagonals.	5.5.4.4	A.3.3.1.5	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COMPACT MEMBERS: All column and brace elements meet section requirements in accordance with AISC 341, Table D1.1a, for highly ductile members. Braced frame beams meet the requirements for moderately ductile members.	5.5.4.3	A.3.3.1.7	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	NET AREA: The brace effective net area is not less than the brace gross area for hollow structural section (HSS) tube and pipe sections.	5.5.4.1	A.3.3.1.8	

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Connections						
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation, and the anchorage is able to develop the least of the following: the tensile capacity of the column, the tensile capacity of the lowest-level column splice (if any), or the uplift capacity of the foundation.	5.7.3.1	A.5.3.1

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